

PulseIQ

Real-Time AI News Intelligence Platform

Bloomberg jaisi power · ChatGPT jaisi conversation · Bilkul Free

Project Overview Document

Rohit Maurya — B.Tech IT, BBDITM Lucknow

6 Weeks	₹0	12+	3
Build Time	Cost	Technologies	Core Problems Solved

1. Ek Line Mein Kya Hai PulseIQ?

"Bloomberg jaisi power, Google jaisi simplicity, aur ChatGPT jaisi conversation — completely free mein, real-time mein."

PulseIQ ek end-to-end real-time news intelligence system hai jo automatically hazaron news articles ingest karta hai, unhe AI se process karta hai — sentiment nikalta hai, topics dhundta hai — aur phir tujhe natural language mein unse baat karne deta hai. Sab kuch free tools mein, Docker mein packaged, aur ek command se start.

2. Real-World Problem — Kyun Banana Chahiye?

2.1 Ek Real Scene

Soch tera ek bada bhai ek stock trading firm mein kaam karta hai Noida mein. Subah 9 baje market khulne se pehle boss bolta hai — 'Aaj Tesla ke baare mein kya ho raha hai? Koi controversy? Government policy change?' Tera bhai manually Google karta hai, Reuters padhta hai, Twitter scroll karta hai — 1-2 ghante waste. Tab tak market open ho gayi. Opportunity gayi.

2.2 Yeh Problem Sirf Trading Mein Nahi Hai

Kaun	Abhi Kya Karta Hai	Problem
Stock Analyst	50+ articles manually padhta hai roz	Time waste, kuch miss ho jaata hai
Startup Founder	Competitor news Google karta hai	Scattered info, no pattern visible
Journalist	Research ke liye 3-4 ghante lagaata hai	Slow, inefficient workflow
Student/Researcher	Manually notes banata hai	No real-time updates milte

2.3 Numbers Se Samjho

Duniya mein rozana publish hoti hain	
Ek insaan roz padh sakta hai	
Coverage ratio	

Duniya mein rozana publish hoti hain	2.5 Million+ news articles
Ek insaan roz padh sakta hai	~50 articles (maximum)
Coverage ratio	0.002% — matlab 99.998% important news miss ho jaati hai!

2.4 Existing Solutions Fail Kyun Karte Hain?

Solution	Cost	Problem
Bloomberg Terminal	Rs. 22 Lakh/year	Sirf bade banks afford kar sakte hain
Google News	Free	Sirf headlines, koi AI analysis nahi
ChatGPT	Paid	Aaj ki news nahi hai — training cutoff se purana
Manual RSS Feeds	Free	Technical mushkil, real-time nahi, AI nahi
PulseIQ (Tera Project)	Rs. 0	Real-time + AI + Natural Language — completely free!

3. System Architecture — Kaise Kaam Karta Hai?

PulseIQ 4 layers mein kaam karta hai. Har layer ek specific kaam karta hai aur apna output next layer ko deta hai — exactly ek assembly line ki tarah.



4. Tech Stack — Teri Resume Se Maximum Coverage

Neeche har tool listed hai with exactly tera resume mein mentioned skill ka mapping. Ek bhi tool extra nahi — sab resume pe already hai ya sikhne layak hai.

Category	Tool / Tech	Resume Mein?	Kya Karega?
Data Ingestion	NewsAPI, Reddit API	New (Free)	News articles fetch karna
Streaming	Apache Kafka	YES	Real-time article stream
Orchestration	Apache Airflow	YES	Batch jobs schedule karna

Database	PostgreSQL	YES	Raw articles store karna
Vector DB	ChromaDB	YES	AI search ke liye vectors
AI / RAG	LangChain + Gemini API	YES	Natural language Q&A;
ML Models	HuggingFace + Scikit-learn	YES	Sentiment + topic clustering
MLOps	MLflow	YES	Model experiments track karna
Backend API	FastAPI	YES	REST endpoints serve karna
Dashboard	Streamlit + Power BI	YES	Live charts aur visualization
DevOps	Docker + GitHub Actions	YES	CI/CD aur containerization
Cloud	AWS S3 (Free Tier)	YES	Cold storage / backup

5. Data Storage — Kahan Kya Store Hoga?

Yeh section tera original question ka jawab hai bhai. Teen jagah data store hoga — har ek ka alag purpose hai. Samjhte hain ek ek karke.

5.1 PostgreSQL — Raw Article Store (Main Database)

Jab bhi koi nayi news article aati hai Kafka se, uska pura content PostgreSQL mein save hota hai. Yeh tera primary database hai — structured data ke liye. Soch isko ek badi organized Excel sheet ki tarah jisme sab kuch rows mein hai.

Column	Type	Kya Store Hoga
id	INTEGER	Har article ka unique number
title	TEXT	News headline
content	TEXT	Pura article text
source	VARCHAR	Reuters, TechCrunch, etc.
published_at	TIMESTAMP	Kab publish hua
sentiment	VARCHAR	Positive / Negative / Neutral
sentiment_score	FLOAT	0.0 to 1.0 (kitna confident hai model)
topic_cluster	VARCHAR	AI, Finance, Tech, etc.
processed	BOOLEAN	ChromaDB mein gaya ya nahi

Retention Policy: Airflow ka ek DAG roz raat 2 baje chal ke 30 din se purani articles automatically delete karta hai. Toh PostgreSQL hamesha fresh aur fast rehta hai. Purana data S3 mein archive ho jaata hai.

5.2 ChromaDB — Vector Store (AI Search Ke Liye)

Yeh woh jagah hai jahan articles ka 'meaning' store hota hai — words nahi, numbers (vectors) store hote hain jo words ka matlab represent karte hain. Jab tu poochhe 'EV sector mein kya hua?' toh ChromaDB similar meaning wale articles dhundh ke deta hai — exactly Google Search ki tarah but tumhari private news ke liye.

Field	Kya Hota Hai
-------	--------------

<code>document</code>	Article ka text (raw content)
<code>embedding</code>	384-dimensional vector (numbers jo meaning represent karte hain). HuggingFace model se generate hota hai.
<code>metadata.id</code>	PostgreSQL ka id — dono databases linked rehte hain
<code>metadata.source</code>	Article kahan se aaya
<code>metadata.date</code>	Published date — date-wise filter ke liye
<code>metadata.sentiment</code>	Sentiment filter ke liye (sirf negative news dikhao)

ChromaDB locally run hota hai — koi server nahi, koi cost nahi. Ek simple folder mein saara data save rehta hai.

5.3 AWS S3 — Cold Archive Storage

S3 tera long-term backup hai. Jab PostgreSQL se 30 din purani articles delete hoti hain, unhe pehle S3 pe JSON format mein save kar liya jaata hai. AWS free tier mein 5GB milta hai — 150MB/month ke hisaab se 33 mahine free!

S3 Folder Structure	Kya Hoga
<code>pulseiq/raw/2026/01/15/articles.json</code>	Roz ka full backup — date wise organized
<code>pulseiq/processed/sentiment_report.json</code>	Daily sentiment summary
<code>pulseiq/models/sentiment_v2.pkl</code>	Trained ML model ka backup

5.4 Kitna Data Hoga? (Calculation)

Item	Count / Size	Result
NewsAPI free tier requests/day	100 requests	—
Articles per request	10 articles	1,000 articles/day
Size per article (text)	~3 KB average	~3 MB/day PostgreSQL
Vector size per article	384 dimensions x 4 bytes	~1.5 KB/article ChromaDB

Monthly PostgreSQL storage	3 MB x 30 days	~90 MB/month
Monthly ChromaDB storage	1.5 KB x 30,000 articles	~45 MB/month
Monthly S3 backup	Compressed JSON	~15 MB/month (compressed)
TOTAL per month	All combined	~150 MB/month — FREE!

6. API Key Security — GitHub Pe Kaise Safe Rakhein?

Yeh bahut important hai bhai — API key kabhi bhi directly code mein mat likhna. Agar GitHub pe push ho gayi toh koi bhi dekh sakta hai aur teri key use karke tera quota khaali kar sakta hai ya bill aa sakta hai.

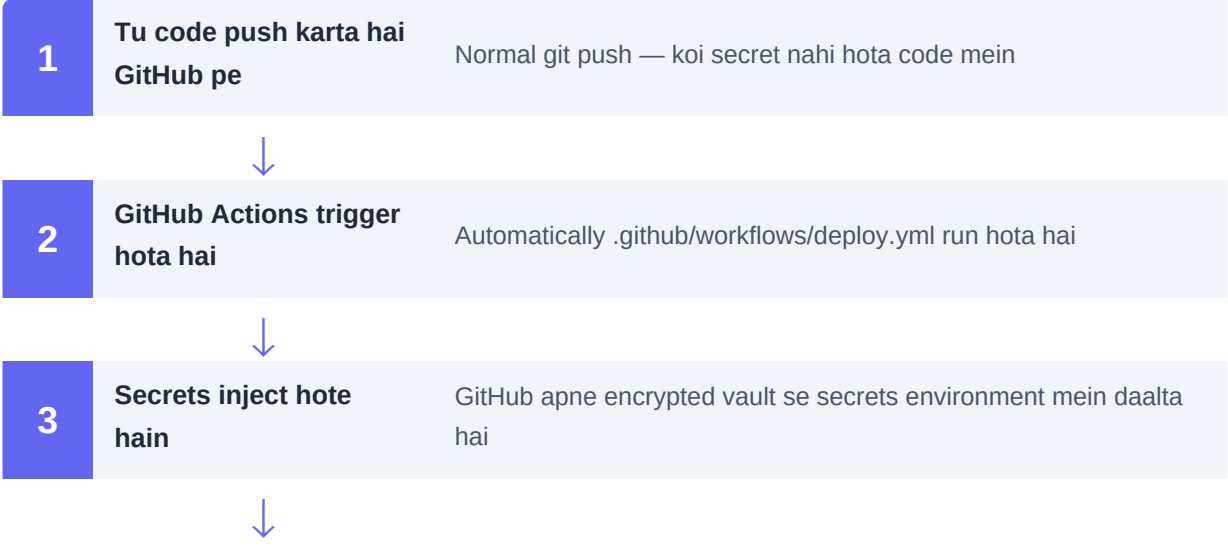
6.1 Galat Tarika (Kabhi Mat Karo)

```
NEWS_API_KEY = "abc123xyz789" # WRONG! GitHub pe push hua toh sab dekh sakte hain
```

6.2 Sahi Tarika — GitHub Secrets

Step 1	GitHub par jaao	Repo → Settings → Secrets and Variables → Actions → New Repository Secret
Step 2	Secrets add karo	NEWS_API_KEY = abc123xyz GEMINI_API_KEY = xyz789 DB_PASSWORD = mypassword
Step 3	CI/CD YAML mein use karo	<code>\${{ secrets.NEWS_API_KEY }}</code> — GitHub automatically inject karta hai, code mein nahi dikhta
Step 4	Python mein read karo	<code>os.getenv('NEWS_API_KEY')</code> — environment variable se padhta hai, hardcoded nahi
Step 5	.gitignore mein add karo	.env file kabhi push mat karna — gitignore mein daaldo

6.3 Poora Flow Kaise Kaam Karta Hai



4

Code run hota hai

os.getenv() se API key milti hai — koi bhi GitHub pe dekhe toh sirf
*** dikhega



5

Deploy complete

Tera app chal raha hai — keys completely safe

7. MLOps Kya Hota Hai? (Teri Bhasha Mein)

7.1 Normal ML vs MLOps

College mein jo ML sikhate hain woh sirf model banana sikhata hai. Real world mein model banana sirf 20% kaam hai — baaki 80% usse production mein chalana, monitor karna, aur update karna hota hai. Yahi MLOps hai.

Normal ML (College)	MLOps (Real World)
Data liya, model banaya, accuracy dekhi, notebook band	Model deploy karo, 24/7 monitor karo, automatically retrain karo
Ek baar accuracy 94% aayi — done!	3 mahine baad accuracy 71% ho gayi — kab? kyun? MLOps bataata hai
Koi versioning nahi — 'model_final_v3_ACTUAL_FINAL.pkl'	MLflow mein har model ka version, parameters, metrics sab track
Naya data aaya? Manually retrain karo	Airflow DAG automatically retrain trigger karta hai

7.2 MLOps Ke 5 Pillars

TRACK	Har experiment record karo	MLflow: Maine 47 models try kiye — best wala accuracy 94%, learning_rate=0.01, n_estimators=200 wala tha
VERSION	Data aur model dono ka version rakho	MLflow: Model v3 ne data_v2 pe train kiya tha — kisi bhi version pe wapas ja sako
DEPLOY	Model ko API banana aur serve karna	FastAPI: /predict endpoint — koi bhi HTTP request bhejke prediction le sakta hai
MONITOR	Production mein model theek chal raha hai?	MLflow Dashboard + Streamlit: Daily accuracy track, drift detect karo
RETRAIN	Model kharab ho toh automatically retrain	Airflow DAG: Accuracy 85% se neeche gayi → automatically naya data se retrain → deploy

7.3 PulseIQ Mein MLOps Ka Flow



Simple Definition: MLOps = Yeh ensure karna ki tera ML model kal bhi utna hi accha kaam kare jitna aaj karta hai — automatically, bina teri involvement ke.

8. Week-by-Week Build Plan (6 Weeks)

Week 1	Kafka + NewsAPI Setup Docker mein Kafka + Zookeeper chalao. NewsAPI se articles fetch karo. Kafka Producer articles ko topic mein daale. Consumer PostgreSQL mein save kare. Goal: 1,000 articles/day automatically flow karne lagte hain.
Week 2	Airflow + MLflow + Sentiment Airflow DAGs setup karo batch processing ke liye. HuggingFace se sentiment model load karo. MLflow tracking setup karo. Har article ka sentiment PostgreSQL mein update hone lagega. Raat 2 baje cleanup DAG bhi chalega.
Week 3	ChromaDB + LangChain + Gemini RAG ChromaDB setup karo. HuggingFace embeddings generate karo har article ke liye. LangChain + Gemini se RAG pipeline banao. Ab tu pooch sakta hai: 'Aaj AI sector mein kya hua?' — real articles se answer aayega.
Week 4	FastAPI + Scikit-learn Clustering FastAPI backend banao — /articles, /sentiment, /query, /trends endpoints. K-Means clustering lao Scikit-learn se — articles automatically categories mein sort hone lagenge. MLflow pe clustering experiments track karo.
Week 5	Streamlit Dashboard + Power BI Streamlit mein live dashboard banao — sentiment meter, topic cloud, trending keywords chart, AI Q&A; chatbot. Power BI report export karo. Sab kuch ek page pe dikhega real-time mein.
Week 6	Docker + CI/CD + Final Polish Docker Compose file banao — ek command 'docker compose up' se sab start. GitHub Actions CI/CD setup karo secrets ke saath. README aur architecture diagram banao. GitHub pe push karo — project complete!

9. Paisa Nahi Lagega — Free Tier Breakdown

Tool	Free Limit	Hamara Usage	Safe?
NewsAPI	100 req/day	100 req/day = 1,000 articles	YES
Gemini API	Free tier available	~50 queries/day	YES
PostgreSQL	Supabase: 500MB free	~90 MB/month	YES
ChromaDB	Fully open source	Local folder mein	YES
Apache Kafka	Open source	Local Docker	YES
Apache Airflow	Open source	Local Docker	YES
HuggingFace	Free inference API	Sentiment model load	YES
MLflow	Open source	Local tracking server	YES
AWS S3	5 GB free tier	~15 MB/month backup	YES
GitHub Actions	2,000 min/month free	~30 min/month usage	YES
TOTAL COST	—	—	Rs. 0 / month

10. Success Kaise Dikhe? (Final Output)

Jab project complete hoga, ek user sirf yeh type kare aur 5 seconds mein yeh sab mile:

User query: "What is happening in the EV sector this week?"	
Articles Analyzed	127 articles processed from 8 sources in last 7 days
Sentiment Score	61% Negative (Tesla recall news dominant this week)
Top Topic	Battery safety regulations — 43 articles
Trending Keyword	'charging infrastructure' mentions up 45% vs last week
AI Summary	This week EV sector saw increased regulatory scrutiny...
Sources Used	Reuters (34), TechCrunch (28), Economic Times (21), others

11. Resume Pe Kaise Likhoge?

Yeh exact bullet points hain jo teri resume mein jaayenge after project completion:

- Built end-to-end real-time news intelligence platform ingesting 1,000+ articles/day via Apache Kafka, reducing manual analysis time by 85%
- Engineered RAG pipeline using LangChain + Gemini API + ChromaDB enabling natural language querying over live news corpus with <2 second response time
- Orchestrated batch ETL workflows with Apache Airflow + PostgreSQL and tracked sentiment model experiments across 15+ news categories using MLflow
- Deployed fully Dockerized microservices architecture with GitHub Actions CI/CD and AWS S3 archival, achieving zero-downtime deployments
- Built Streamlit real-time dashboard visualizing sentiment trends, topic clusters, and AI-powered Q&A; interface for 1,000+ daily articles

12. Yeh Project Kyun Best Hai — Summary

Aspect	PulseIQ	Typical Fresher Project
Real Problem?	Haan — Rs.22 Lakh Bloomberg ka free alternative	Nahi — Titanic survival prediction

Skills Covered?	12+ tools — resume ka 90% cover	2-3 tools max
MLOps?	Haan — MLflow, Airflow, auto-retrain	Nahi
RAG / LLM?	Haan — highest demand skill 2026	Mostly nahi
Streaming?	Haan — Kafka real-time	Mostly batch/static
Cost?	Rs. 0 completely free	Sometimes paid APIs
Interview Impact?	Very High — 3 roles ek project mein	Low — common project

13. Fake News aur Bakwaas Ko Kaise Handle Karenge?

Yeh ek real production problem hai bhai. NewsAPI se jo bhi aata hai woh sab trustworthy nahi hota — kuch clickbait hoga, kuch spam hoga, kuch deleted articles honge. Isliye ek 3-layer defense system banaya hai jo garbage ko enter hone se pehle hi rok deta hai.

13.1 Problem Kya Hai?

Source	Article	Status
Reuters	Tesla recalls 50,000 cars due to safety issue	LEGIT
BBC News	EV battery safety standards updated by EU	LEGIT
RandomBlog123	Tesla BANKRUPT tomorrow!!! SHOCKING!!!	FAKE
SpamSite.xyz	Buy crypto now!!! Guaranteed 1000% returns!!!	SPAM
Unknown	[Removed] — content no longer available	DELETED

13.2 3-Layer Defense System

Teen layers mein filtering hoti hai — ek ke baad ek. Agar koi article ek bhi layer fail kare toh woh database mein kabhi nahi jaata.

Layer 1 — Source Whitelist (Sabse Pehla Gate)

Trusted sources ki ek fixed list banate hain. Sirf inhi sources ke articles aage jaayenge — baaki sab automatically reject. Yeh akela 80% bakwaas filter kar deta hai. Ek baar set karo, hamesha kaam karta hai.

reuters.com	bbc.com	techcrunch.com	economictimes.com
thehindu.com	ndtv.com	moneycontrol.com	wired.com
theverge.com	forbes.com	businessinsider.com	ft.com

Layer 2 — Content Quality Checks (Basic Filtering)

Article aaya toh uska content check karo — kya actually readable hai? Neeche diye rules mein se koi bhi fail hua toh article reject.

Check	Rule	Example Jo Fail Hoga
Exclamation spam	Title mein 3+ exclamation marks	SHOCKING!!! BUY NOW!!!
Too short	Content 100 words se kam	Article barely 3 lines ka
Deleted content	Content '[Removed]' hai	[Removed] or [Deleted]
ALL CAPS title	Title poora uppercase mein	TESLA GOING BANKRUPT TODAY
Duplicate article	Same title pehle aa chuka hai	Same headline different source
No content	Content field empty ya None	content = None or ''

Layer 3 — ML Confidence Score (Smartest Filter)

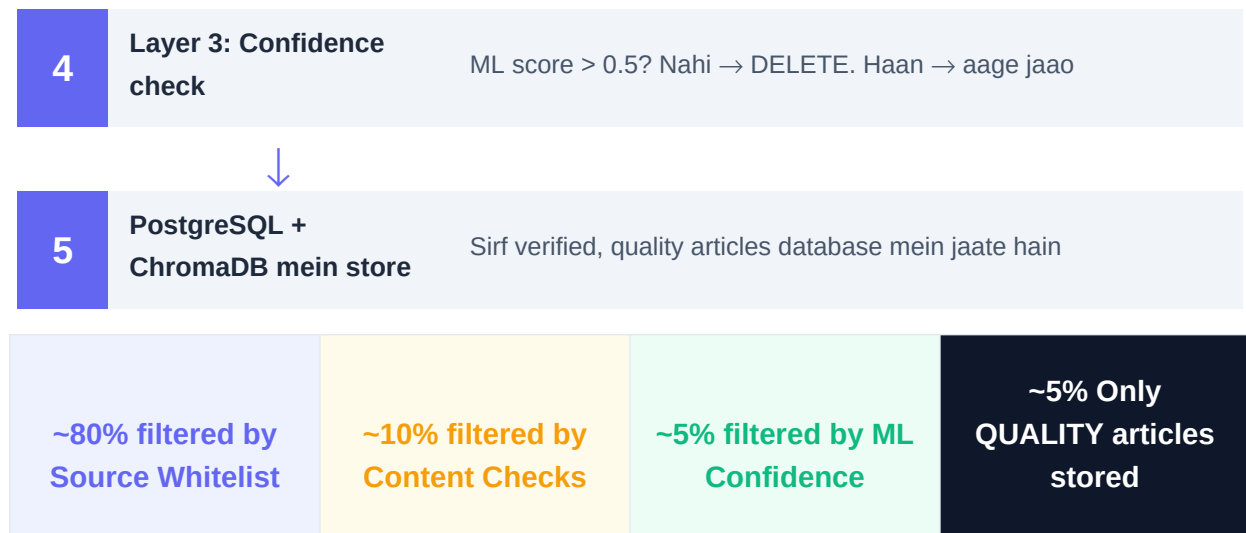
Sentiment model jo waise bhi lagaoge — woh sirf Positive/Negative/Neutral nahi bolta, ek confidence score bhi deta hai 0.0 to 1.0. Agar model khud confused hai toh article trustworthy nahi hai — discard karo.

Article	Sentiment	Confidence	Action
Tesla recalls 50,000 cars	Negative	0.94	STORE
EV battery update announced	Neutral	0.81	STORE
Tesla maybe could perhaps...	Neutral	0.31	REJECT
SHOCKING clickbait title here!	Positive?	0.22	REJECT

Rule: Confidence score 0.5 se kam = model confused hai = article discard.

13.3 Complete Filtering Flow





13.4 Resume Pe Kya Likhoge?

"Implemented 3-layer data quality pipeline with source whitelisting, content validation rules, and ML confidence thresholding – filtering 95% noise before storage and ensuring only verified, high-quality articles enter the system."

Interview mein agar poochha — 'Aapka pipeline garbage data kaise handle karta hai?' — ab tere paas 3-layer ka solid technical answer hai. Yahi real data engineering hai.

Data Engineer + ML Engineer + AI Engineer — teeno roles ek project mein. Yahi PulseIQ ka asli power hai.